Gravimetric verification at the IV workflow cutover: batch-compounded products

To: Cynthia Underwood, Director of Pharmacy. From: Erin Stewart, quality pharmacist, Sterile Compounding Unit. Cc: Spencer Frank, IV Workflow Project Lead. 8 February 2027.

Recommendation

Configure 4 batch-compounded products for gravimetric verification of every dose at the 5 April 2027 cutover: Cefoxitin 2 g in 20 mL syringe, Phenylephrine 100 mcg/mL 10 mL syringe, Cefazolin 2 g in 20 mL syringe and Midazolam 1 mg/mL 50 mL syringe. Each made more than 10 out-of-tolerance doses per 100,000 doses in 2026 (PH-QP-07 3.1). The other 8 batch products stay on the in-process checks of SC-SOP-12; potassium chloride and norepinephrine stay on GC-1 (3.2).

How the 2026 rate was estimated

PH-QP-07 counts every dose made outside tolerance, weighed or not, and SC-SOP-12 weighs the first dose of a batch and then one syringe in a hundred or one bag in twenty, so the rate has to come from a model of the delivered volume. On the peristaltic stations each set delivers less the more it has pumped: across every set, a product's weighed doses fall in a straight line with the litres the set had pumped since its set-up, about a start that varies from set to set. For each product we fitted that start and the loss per litre on the weighed doses (the check register tied to each dose in the server's dose log, each dose placed on its set from the compounders' set-up events), took every completed dose at the litres its set had pumped, and added up each dose's chance of lying outside the product's tolerance.

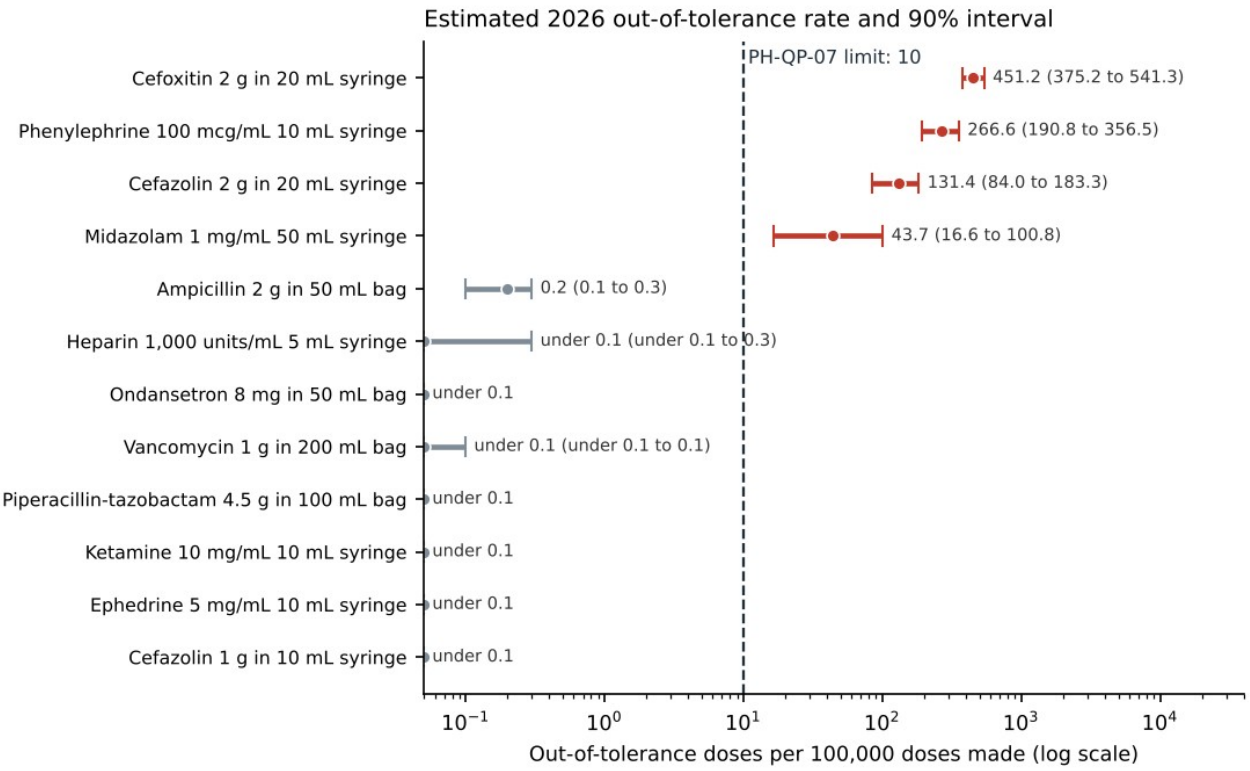
The 90 per cent interval on each rate comes from refitting the line on the product's sets drawn again with replacement, a thousand times, and adding up its 2026 doses each time. The limit sits below the interval of every product on the list (midazolam's lower bound is 16.6) and above the interval of every other product (the highest upper bound is 0.3), so the list does not turn on where in its interval a rate falls.

The rate is driven by the sets that carried extra batches. Cefazolin 2 g and phenylephrine sets that ran three batches ended near the bottom of the band after the batch's last weighed dose, so almost none of their low doses were weighed and no quarantine sort was opened on either. Cefoxitin's large batches cross the limit inside a second batch and its checks caught part of them. Midazolam does not move on PC-4 but sits 2.6 per cent under target, so a small share of every batch falls outside.

A normal fit to each product's weighed doses (the usual capability figure) would put ampicillin, piperacillin-tazobactam and vancomycin bags on the list: their sets start 2.2 to 2.4 per cent high and lose about 4.1 per cent over a batch, so the weights spread wide, but no set comes near the limit before it is changed.

Product	Station	Doses made, 2026	Out-of-tolerance rate (per 100,000)	90% interval	Limit of 10	PH-QP-07 3.1
Cefoxitin 2 g in 20 mL syringe	PC-1	96,015	451.2	375.2 to 541.3	Below the interval	In: verify every dose
Phenylephrine 100 mcg/mL 10 mL syringe	PC-2	89,743	266.6	190.8 to 356.5	Below the interval	In: verify every dose
Cefazolin 2 g in 20 mL syringe	PC-1	90,478	131.4	84.0 to 183.3	Below the interval	In: verify every dose
Midazolam 1 mg/mL 50 mL syringe	PC-4	18,770	43.7	16.6 to 100.8	Below the interval	In: verify every dose
Ampicillin 2 g in 50 mL bag	PC-3	22,313	0.2	0.1 to 0.3	Above the interval	Out: SC-SOP-12 checks
Cefazolin 1 g in 10 mL syringe	PC-1	26,723	under 0.1	under 0.1	Above the interval	Out: SC-SOP-12 checks
Ephedrine 5 mg/mL 10 mL syringe	PC-2	30,852	under 0.1	under 0.1	Above the interval	Out: SC-SOP-12 checks
Ketamine 10 mg/mL 10 mL syringe	PC-2	17,006	under 0.1	under 0.1	Above the interval	Out: SC-SOP-12 checks
Piperacillin-tazobactam 4.5 g	PC-3	22,266	under 0.1	under 0.1	Above the	Out: SC-

in 100 mL bag					interval	SOP-12 checks
Vancomycin 1 g in 200 mL bag	PC-3	7,266	under 0.1	under 0.1 to 0.1	Above the interval	Out: SC-SOP-12 checks
Ondansetron 8 mg in 50 mL bag	PC-3	8,534	under 0.1	under 0.1	Above the interval	Out: SC-SOP-12 checks
Heparin 1,000 units/mL 5 mL syringe	PC-4	26,248	under 0.1	under 0.1 to 0.3	Above the interval	Out: SC-SOP-12 checks



Sources: compounder dose log and event export (pharmacy server, exported 18 January 2027); gravimetric check register and quarantine register, 2026; compounded product master; equipment register. Weights converted to volume with each product's specific gravity; gross weights before 1 July less the container tare.